



1.5

Polynomial Functions and Complex Zeros

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1

| ECHS Mathematics

| Teacher: Mohammad Abu-

Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: A degree- n polynomial has exactly n complex zeros counting multiplicity.

2. Write one precise sentence explaining this principle: Nonreal zeros of a real-coefficient polynomial occur in conjugate pairs.

3. Write one precise sentence explaining this principle: Odd multiplicity usually produces crossing at a real zero; even multiplicity produces touching.

4. Write one precise sentence explaining this principle: A zero r corresponds to a factor $x - r$.

Core

5. A real quartic has zeros 2 and $1 + 3i$. List all zeros if 2 has multiplicity 2.

6. Construct a monic real cubic with zeros -1 and $2 + i$.

7. At $x = 3$, a graph flattens and crosses. What multiplicity is plausible?

8. State what the formula $P(r) = 0 \iff (x - r)$ is a factor tells you and name any required restriction.

9. State what the formula $z = a + bi \Rightarrow \bar{z} = a - bi$ is also a zero tells you and name any required restriction.

10. State what the formula $\sum \text{multiplicities} = \deg P$ tells you and name any required restriction.

Medium

11. Analyze zeros and multiplicities for $P(x) = (x + 2)^2(x - 1)^3$.

12. Analyze zeros and multiplicities for $P(x) = x^4 - 5x^2 + 4$.

13. Analyze zeros and multiplicities for $P(x) = x^3 + 4x$.

14. Analyze zeros and multiplicities for $P(x) = x^4 + 4$.

15. Analyze zeros and multiplicities for $P(x) = (x - 3)^4(x + 1)$.

Hard

16. A real polynomial of degree 6 has zeros 2 (multiplicity 2), -1 , and $3 + 4i$. What zeros are still required and how many degrees remain?

17. Construct the least-degree monic real polynomial with zeros -2 , 1 of multiplicity 2, and $2 - i$.

18. Determine k so $x = 2$ is a zero of $P(x) = x^3 - kx^2 + 4$.

19. A polynomial graph crosses at $x = -1$, touches at $x = 2$, and has degree 5. Give a possible factored form with positive leading coefficient.

Difficult

20. A real quartic has $1 + i$ as a double zero. Find the polynomial up to a nonzero real constant.

21. Find all real k for which $x^2 + kx + 9$ has a repeated real zero.

22. Prove that a real polynomial of odd degree has at least one real zero.

Challenge / HOT

23. Can a real degree-7 polynomial have exactly two distinct real zeros, both of even multiplicity? Explain.

24. Construct two degree-6 real polynomials with the same real zeros and multiplicities but different complex zeros.

AP-Style Multiple Choice

25. A nonreal zero $2 + 3i$ of a real polynomial forces

- (A) $-2 + 3i$
- (B) $2 - 3i$
- (C) $-2 - 3i$
- (D) $3 + 2i$

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.